Huawei Access Node HEB-ACC-42 Hebbal Edge Hardware/Environmental Fault Resolution Procedure (ALARM_8682)

Node: HEB-ACC-42 | **Vendor:** Huawei | **Severity:** Major | **Alarm Code:** ALARM_8682

Description

This SOP outlines the step-by-step procedure to diagnose and resolve a hardware/environmental fault on the Huawei Access Node HEB-ACC-42 at the Hebbal Edge site, resulting in a Major Alarm (ALARM_8682) due to ASIC memory parity errors, line card crashes, or power supply redundancy loss. The Huawei Access Node is a critical component of the Hebbal Edge site's access network, providing connectivity to downstream routers and switches. The access node's hardware and environmental conditions are subject to various failure modes, including hardware malfunctions and environmental factors such as temperature, humidity, and power supply quality. In this scenario, the failure domain is Hardware/Environmental Fault, indicating that the issue is likely related to a hardware or environmental condition rather than a software or configuration issue. The alarm code ALARM_8682 indicates a Major alarm, which requires immediate attention to prevent network downtime and potential data loss.

Pre-requisite Checks

- Confirm the site location and node ID to ensure the correct node is identified.

- Verify the node's power supply redundancy status to ensure the node has sufficient power supply backup.

- Check the node's environmental conditions, including temperature, humidity, and airflow, to ensure they are within the recommended operating ranges.

- Dispatch remote hands to the Hebbal Edge site to assist with physical access and troubleshooting.

Potential Root Causes

Cause	Probability
ASIC memory parity errors due to faulty memory modules or high temperature	High
Line card crashes due to software bugs or hardware malfunctions	Medium
Power supply redundancy loss due to faulty power supplies or high current draw	Medium

Diagnostic Commands & Expected Outputs

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Command: display device internal info, Expected Output: Device internal information, including temperature, voltage, and current values, Description: This command displays the internal information of the device, including temperature, voltage, and current values, which can help identify potential hardware issues.
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Command: display power supply info, Expected Output: Power supply information, including status, voltage, and current values, Description: This command displays the power supply information, including status, voltage, and current values, which can help identify potential power supply issues.

Command: debug linecard crash, Expected Output: Line card crash debug information, Description: This command enables line card crash debugging, which can provide detailed information about the line card crash, including the cause and affected interfaces.
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Resolution Steps

Step 1: Step: Replace faulty memory modules or line cards, Command: replace device memory, Description: Replace faulty memory modules or line cards with new ones to resolve the issue.

Step 2: Step: Reboot the node, Command: reboot device, Description: Reboot the node to ensure all changes take effect and the issue is resolved.

Step 3: Step: Verify power supply redundancy, Command: display power supply info, Description: Verify the power supply redundancy status to ensure the node has sufficient power supply backup.

Post-Check Verification

```
Command: display device internal info, Expected Output: Device internal information, including temperature, voltage, and current values, Description: Verify the device internal information to ensure the issue is resolved.

Command: display power supply info, Expected Output: Power supply information, including status, voltage, and current values, Description: Verify the power supply information to ensure the issue is resolved.
```

Roll-back Procedures

- Step: Restore the original configuration, Command: restore configuration, Description: Restore the original configuration to the node to revert changes and resolve the issue.

- Step: Disable debugging, Command: undo debug linecard crash, Description: Disable debugging to prevent further logging and resolve the issue.